

Announcements

► Any questions/issues?

► Schedule:

Today (Follow-up): Discussion Q2 (absorption lines);
Discussion Q3 (**now a bonus question**)

Today (Follow-up): Boiling Lab — what did we observe and what does it mean?

Today (Follow-up): Where are we on T_{Earth} ?

Today (New): Why is there evaporation below boiling?

Tomorrow (Lab): Evaporation of water and clouds!

Thursday (Discussion): Final calculation of Earth's equilibrium temperature.

Next Week: No Lab; Starting paleo perturbations of temperature.

Today's Reading Quiz (3 min)

On your own sheet of paper. . .

Say a few words about each: *atmospheric pressure*,
humidity of air, *latent heat*.

Turn in to Blackboard as scanned pdf (or image is fine).

LET'S TALK A BIT ABOUT ATMOSPHERIC PRESSURE

- ② READING MENTIONED THAT THE WEIGHT OF A $1\text{ cm} \times 1\text{ cm}$ COLUMN OF AIR (SURFACE TO TOP OF ATMOSPHERE) IS $1,000\text{ g} = 1\text{ kg}$.
SO THIS IS A NICE EXPRESSION OF PRESSURE (FORCE PER AREA):

$$P_{\text{ATM}} = \frac{1\text{ kg}}{1\text{ cm}^2} \times \frac{2.2\text{ lb}}{1\text{ kg}} = 2.2 \frac{\text{lb}}{\text{cm}^2}$$

THAT'S TWO PASTA PACKAGES ($1\text{ lb} = 454\text{ g} = 0.45\text{ kg}$) PRESSING ON EVERY SQUARE CENTIMETER.

- ③ IN OUR STANDARD UNITS:

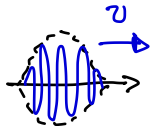
$$\begin{aligned} 2.2 \frac{\text{lb}}{\text{cm}^2} &= 2.2 \frac{\text{lb}}{\text{cm}^2} \times \frac{2.54\text{ cm}}{1\text{ in}} \times \frac{2.54\text{ cm}}{1\text{ in}} \\ &= 14 \frac{\text{lb}}{\text{in}^2} = 14 \text{ "POUNDS PER SQUARE INCH"} \\ &= 14 \text{ PSI} \end{aligned}$$

- ④ YOUR CAR TIRE INFLATES TO 32 PSI, BUT THAT IS THE "GAGE PRESSURE", ADDITIONAL PRESSURE ON TOP OF P_{ATM} ... SO THE TIRE IS 46 PSI RELATIVE TO THE VACUUM OF SPACE.

- ⑤ WHY DON'T WE FEEL THAT PRESSURE ON OUR BODIES? BECAUSE OUR INSIDES ARE ALSO AT ABOUT THE SAME PRESSURE ... WE ARE IN EQUILIBRIUM.

(IF YOU JUMP OUT OF A SPACESHIP, YOU WILL EXPLODE!)

Question 2 from Discussion



$$\lambda = 2.73 \mu\text{m}$$

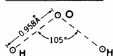
ENERGY OF PHOTON

$$E = hf$$

$$= h \left(\frac{c}{\lambda} \right)$$

$$c = \lambda f$$

Water vapor molecule



Vibratory states



Symmetric band stretching
 $\nu_1 = 3657 \text{ cm}^{-1} = 2.73 \mu\text{m}$



Bending
 $\nu_2 = 1595 \text{ cm}^{-1} = 6.25 \mu\text{m}$



Asymmetric band stretching
 $\nu_3 = 3756 \text{ cm}^{-1} = 2.66 \mu\text{m}$

Ozone molecule



Vibratory states



Symmetric band stretching
 $\nu_1 = 1110 \text{ cm}^{-1} = 9.0 \mu\text{m}$



Bending
 $\nu_2 = 701 \text{ cm}^{-1} = 14.3 \mu\text{m}$

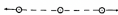


Asymmetric band stretching
 $\nu_3 = 1045 \text{ cm}^{-1} = 9.6 \mu\text{m}$

Carbon dioxide molecule



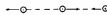
Vibratory states



Symmetric band stretching
No band



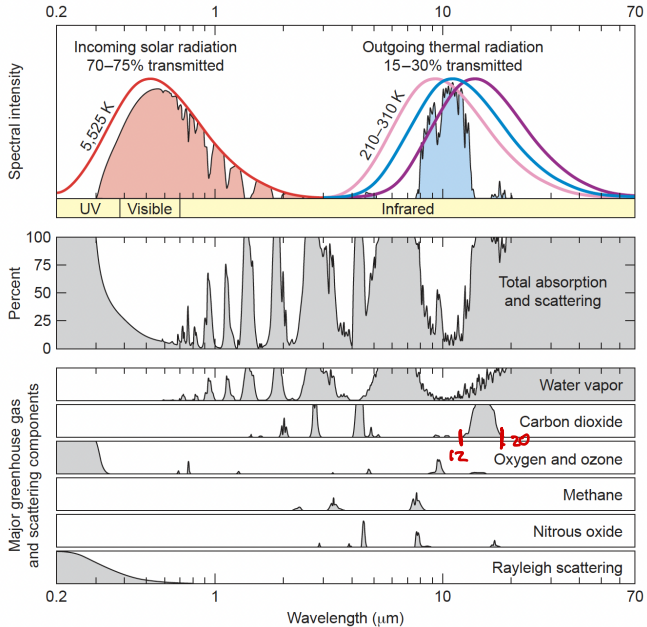
Bending
 $\nu_2 = 667 \text{ cm}^{-1} = 15 \mu\text{m}$



Asymmetric band stretching
 $\nu_3 = 2349 \text{ cm}^{-1} = 4.3 \mu\text{m}$

FIGURE 6.5. The structures of the water vapor, ozone, and carbon dioxide molecules and their vibratory states (from Houghton, 1985).

Question 3 from Discussion



[2]

$T_{\text{Earth}} = 288 \text{ K}$ ($= 15^\circ\text{C} = 60^\circ\text{F}$) — AVERAGE EARTH SURFACE TEMP.

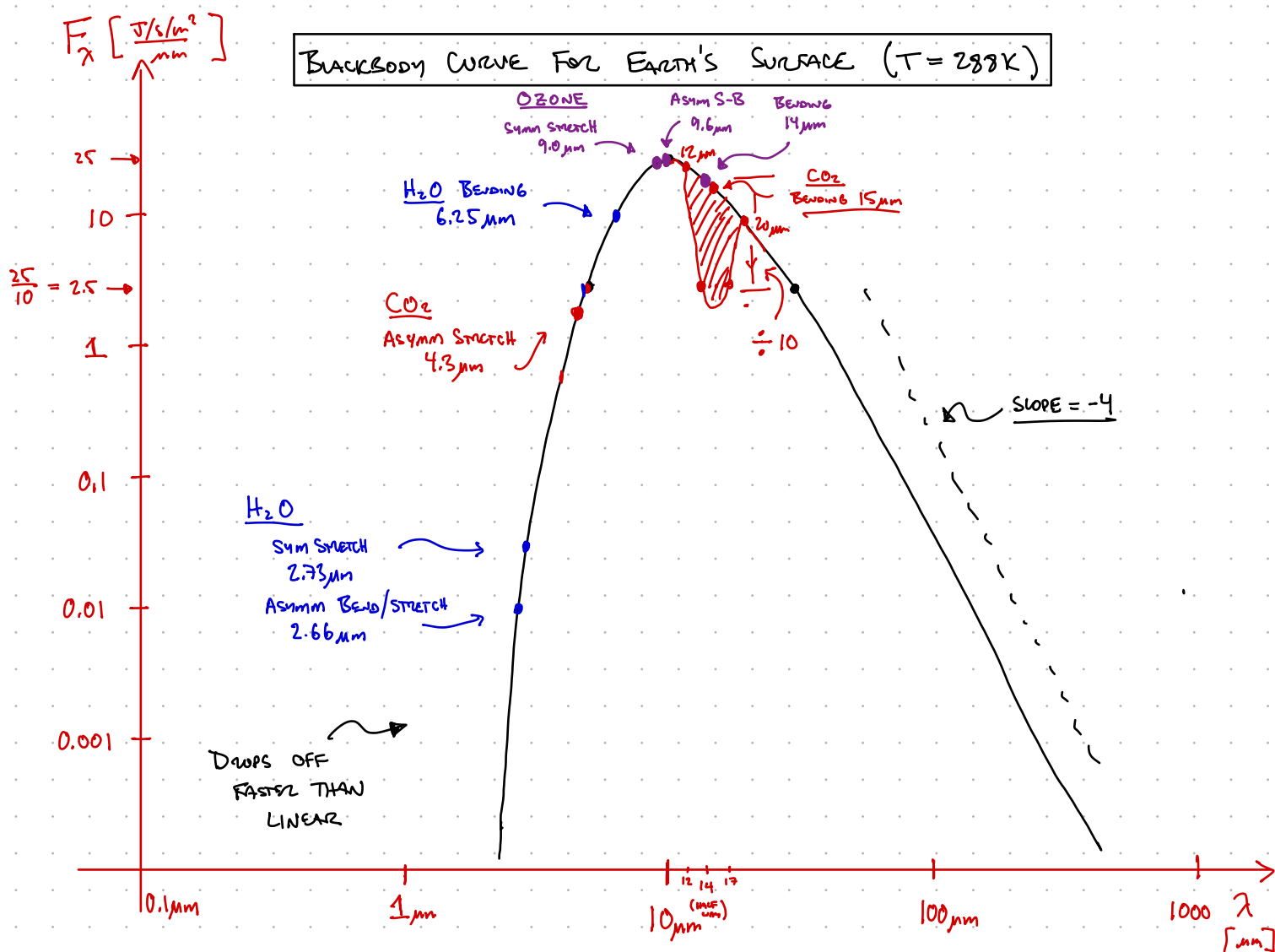
From Discussion 4: BLACKBODY RADIATION

$$\lambda_{\text{peak}} = \frac{2900 \mu\text{m} \cdot \text{K}}{T} = \frac{2900 \mu\text{m} \cdot \text{K}}{288 \text{ K}} = 10.1 \mu\text{m}$$

$$F_{\text{TOT}} = \left(5.7 \times 10^{-8} \frac{\text{J/s}}{\text{m}^2 \text{K}^4} \right) T^4 = 392 \frac{\text{J/s}}{\text{m}^2}$$

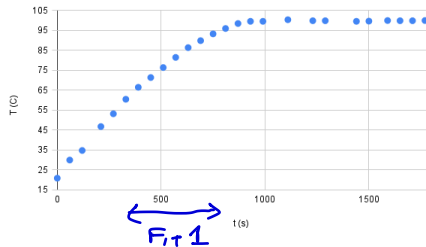
$$F_{\lambda, \text{PEAK}} = \frac{F_{\text{TOT}}}{1.5 \lambda_{\text{peak}}} \approx 26 \frac{\text{J/s/m}^2}{\mu\text{m}}$$

$$\lambda \left(\frac{F_{\lambda}}{10} \right) = \left(\frac{\lambda_{\text{peak}}}{2}, 3 \lambda_{\text{peak}} \right) \approx (5 \mu\text{m}, 30 \mu\text{m})$$

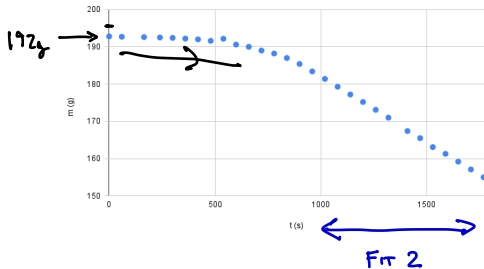


Boiling Lab: Data

T (C) vs. t (s)



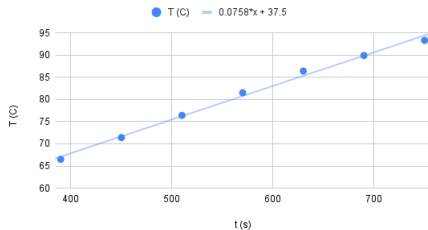
m (g) vs. t (s)



Boiling Lab: Analysis

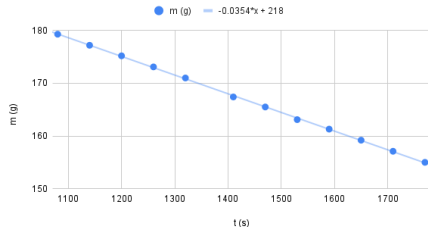
Let's calculate the *latent heat of vaporization*, L_v , knowing that the specific heat capacity of water is 4.2 J/gK .

T (C) vs. t (s)



$$\frac{\Delta T}{\Delta t} = 0.0758 \frac{\text{K}}{\text{s}}$$

m (g) vs. t (s)



$$\left| \frac{\Delta m}{\Delta t} \right| = 0.0354 \frac{\text{g}}{\text{s}}$$

Prior to Boiling

Pick a
 Δt
Time Interval

$$\Delta E_{\text{INT}} = m_w \cdot c_w \cdot \Delta T$$

$$\frac{\Delta E_{\text{INT}}}{\Delta t} = m_w c_w \left(\frac{\Delta T}{\Delta t} \right)$$

Slope 1

Power of
Hotplate
causes $\frac{\Delta E_{\text{INT}}}{\Delta t}$

$$P_{\text{Hotplate}} = m_w c_w \left(\frac{\Delta T}{\Delta t} \right)$$

(T was
const
@ 100°C)

After Boiling

$$\Delta E_{\text{INT}} = m_w \cdot c_w \cdot \cancel{\Delta T}$$

$$+ \Delta m_w L_v$$

Latent
Heat of
Vapor.
($\frac{\text{J}}{\text{kg}}$)

Divide
by
 Δt

$$\frac{\Delta E_{\text{INT}}}{\Delta t} = \left(\frac{\Delta m_w}{\Delta t} \right) L_v$$

Slope 2

Power of
Hotplate

$$P_{\text{Hotplate}} = \left(\frac{\Delta m_w}{\Delta t} \right) L_v$$

$$m_w c_w \left(\frac{\Delta T}{\Delta t} \right) = \left(\frac{\Delta m_w}{\Delta t} \right) L_v$$

$$L_v = \frac{m_w c_w \left(\frac{\Delta T}{\Delta t} \right)}{\left(\frac{\Delta m_w}{\Delta t} \right)} \approx 1800 \frac{\text{J}}{\text{kg}}$$

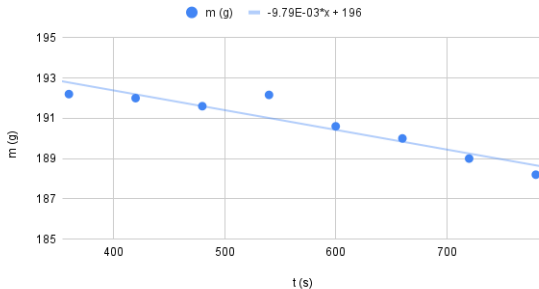
$$(m_w = 192 \text{ g}, c_w = 4.2 \frac{\text{J}}{\text{kg} \cdot \text{K}})$$

Slopes on Graph

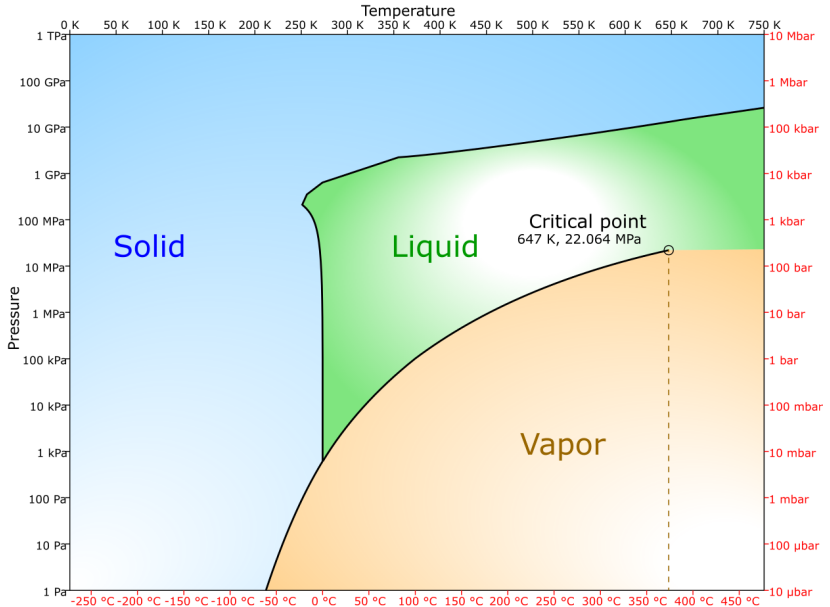
$$\left(\text{Actual is } L_v = 2200 \frac{\text{J}}{\text{kg}} \right)$$

Boiling Lab: Analysis (2nd Attempt)

m (g) vs. t (s)



Phase Transitions in (Pure) Water



Water Evaporation in Air

Consider this figure from the *Mathez* textbook reading today:

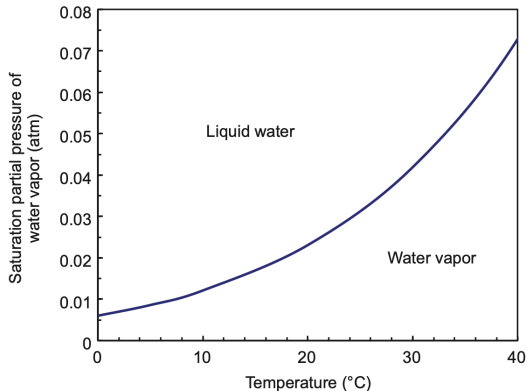


FIGURE 1.2 The water vapor pressure in air in equilibrium with liquid water as a function of temperature at 1 atmosphere total pressure

The curve was calculated from the Clausius-Clapeyron equation.

Why are the values of the pressure so different along the line of transition from liquid to vapor?

Water Evaporation in Air

Answer: Because evaporation in air uses the so-called **partial pressure** of water within the whole air:

$$p_{\text{partial, H}_2\text{O}} = \frac{\text{moles}_{\text{H}_2\text{O}}}{\text{moles}_{\text{air}}} \times p_{\text{atm}}$$

Where $p_{\text{atm}} = 101 \text{ kPa} = 1 \text{ atm}$ is the full atmospheric pressure. Typically the mole fraction above is a few percent. Let's try with 2% ... what is the new boiling point?

Calculating the equilibrium T_{Earth}

Recall our steps to calculate the equilibrium temperature of Earth thus far:

- ▶ Planet with no atmosphere, receiving solar radiation from the sun at $S/4(1 - \alpha)$
- ▶ Planet receiving that solar radiation, with an atmosphere.

Now (in Discussion this week) we will make one final adjustment:

- ▶ Planet with an atmosphere, plus transport of water vapor above the greenhouse gas layer.